ESRTOS Lab Exam

Complete Answer Sheet | AVR-GCC | ATmega

Lab Exam 1

Question 1

Toggle alternate bits (10101010 pattern) on PORTC continuously using XOR.

```
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>

int main(void) {
    DDRC = 0xFF;
    PORTC = 0xAA;
    while(1) {
        PORTC ^= 0xFF;
        _delay_ms(500);
    }
}
```

Question 2

Implement AND, OR and XOR between 0x55 and 0x0F — store result in PORTA.

```
#include <avr/io.h>

int main(void) {
    DDRA = 0xFF;
    unsigned char a = 0x55, b = 0x0F;
    PORTA = a & b;    // AND -> 0x05
    PORTA = a | b;    // OR  -> 0x5F
    PORTA = a ^ b;    // XOR -> 0x5A
    while(1);
}
```

Question 3

Toggle only even bits (0, 2, 4, 6) of PORTB.

```
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>

int main(void) {
    DDRB  = 0xFF;
    PORTB = 0x55;         // 01010101 - even bits HIGH
    while(1) {
        PORTB ^= 0x55;    // toggle bits 0,2,4,6 only
        _delay_ms(500);
    }
}
```

Question 4

Generate values from 0x10 to 0xF0 and display on PORTC with delay.

```
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>

int main(void) {
    DDRC = 0xFF;
    unsigned char i;
    while(1) {
        for(i = 0x10; i <= 0xF0; i += 0x10) {
            PORTC = i;
            _delay_ms(500);
        }
    }
}
```

Question 5

Display the complement of 0xA5 on PORTB.

```
#include <avr/io.h>

int main(void) {
    DDRB  = 0xFF;
    PORTB = ~0xA5;    // result = 0x5A
    while(1);
}
```

Question 6

Convert two ASCII digits ('3' and '7') into packed BCD.

```
#include <avr/io.h>

int main(void) {
    unsigned char ascii1  = '3';
    unsigned char ascii2  = '7';
    unsigned char bcd1    = ascii1 & 0x0F;       // 0x03
    unsigned char bcd2    = ascii2 & 0x0F;       // 0x07
    unsigned char packed  = (bcd1 << 4) | bcd2;  // 0x37
    DDRB  = 0xFF;
    PORTB = packed;
    while(1);
}
```

Question 7

Monitor PB3: if HIGH send 0xFF to PORTD, otherwise send 0x00.

```
#include <avr/io.h>

int main(void) {
    DDRB = 0x00;
    DDRD = 0xFF;
    while(1) {
        if (PINB & (1 << PB3))
            PORTD = 0xFF;
        else
            PORTD = 0x00;
    }
}
```

Lab Exam 2

Question 1

Display "EMBEDDED SYSTEM" on LCD.

```
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>

// Connections: RS=PD0, EN=PD1, D4-D7=PD4-PD7
void lcd_pulse_en(void) {
    PORTD |=  (1<<PD1); _delay_us(1);
    PORTD &= ~(1<<PD1); _delay_us(100);
}
void lcd_nibble(unsigned char n) {
    PORTD = (PORTD & 0x0F) | (n << 4);
    lcd_pulse_en();
}
void lcd_send(unsigned char d, unsigned char rs) {
    if(rs) PORTD |=  (1<<PD0);
    else   PORTD &= ~(1<<PD0);
    lcd_nibble(d >> 4);
    lcd_nibble(d & 0x0F);
    _delay_ms(2);
}
void lcd_init(void) {
    DDRD = 0xFF; _delay_ms(20);
    lcd_nibble(0x03); _delay_ms(5);
    lcd_nibble(0x03); _delay_ms(1);
    lcd_nibble(0x03); _delay_ms(1);
    lcd_nibble(0x02); _delay_ms(1);
    lcd_send(0x28,0); lcd_send(0x0C,0);
    lcd_send(0x01,0); lcd_send(0x06,0);
}
void lcd_print(const char *s) {
    while(*s) lcd_send(*s++, 1);
}
int main(void) {
    lcd_init();
    lcd_print("EMBEDDED SYSTEM");
    while(1);
}
```

Question 2

Blink LED using Timer0 with prescaler 64.

```
#include <avr/io.h>

int main(void) {
    DDRB   = 0xFF;
    TCCR0A = (1<<WGM01);            // CTC mode
    TCCR0B = (1<<CS01)|(1<<CS00);   // prescaler 64
    OCR0A  = 249;                    // 1 ms @16 MHz
    unsigned int count = 0;
    while(1) {
        if (TIFR0 & (1<<OCF0A)) {
            TIFR0 = (1<<OCF0A);
            if (++count >= 500) {
                PORTB ^= (1<<PB0);
                count = 0;
            }
        }
    }
}
```

Question 3

Blink LED connected to PB2 using Timer0 overflow.

```
#include <avr/io.h>
#include <avr/interrupt.h>

volatile unsigned int ovf_count = 0;

ISR(TIMER0_OVF_vect) {
    if (++ovf_count >= 30) {
        PORTB ^= (1<<PB2);
        ovf_count = 0;
    }
}
int main(void) {
    DDRB  |= (1<<PB2);
    TCCR0B = (1<<CS02)|(1<<CS00);   // prescaler 1024
    TIMSK0 = (1<<TOIE0);
    sei();
    while(1);
}
```

Question 4

Division of two variables (dynamic input) — result on PORTB.

```
#include <avr/io.h>

int main(void) {
    DDRB = 0xFF;
    DDRC = 0x00;    // dividend input on PORTC
    DDRD = 0x00;    // divisor  input on PORTD
    unsigned char a = PINC;
    unsigned char b = PIND;
    PORTB = (b != 0) ? (a / b) : 0xFF;   // 0xFF = error
    while(1);
}
```

Question 5

Subtraction of two variables (dynamic input) — result on PORTD.

```
#include <avr/io.h>

int main(void) {
    DDRD = 0xFF;
    DDRC = 0x00;    // a on PORTC
    DDRB = 0x00;    // b on PORTB
    unsigned char a = PINC;
    unsigned char b = PINB;
    PORTD = a - b;
    while(1);
}
```

Question 6

Read BCD value and convert it to ASCII.

```
#include <avr/io.h>

int main(void) {
    DDRC = 0x00;    // packed BCD input on PORTC
    DDRB = 0xFF;
    DDRD = 0xFF;
    unsigned char bcd  = PINC;
    unsigned char high = (bcd >> 4) & 0x0F;
    unsigned char low  =  bcd & 0x0F;
    PORTB = high + '0';    // ASCII tens  digit
    PORTD = low  + '0';    // ASCII units digit
    while(1);
}
```

Question 7

XOR operation between two numbers (dynamic input) — result on PORTB.

```
#include <avr/io.h>

int main(void) {
    DDRC = 0x00;    // input a on PORTC
    DDRD = 0x00;    // input b on PORTD
    DDRB = 0xFF;
    unsigned char a = PINC;
    unsigned char b = PIND;
    PORTB = a ^ b;
    while(1);
}
```